JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 94, NO. B8, PAGES 10,669–10,683, AUGUST 10, 1989

A New Siluro-Devonian Paleopole From Early Paleozoic Rocks of the Franklinian Basin, North Greenland Fold Belt

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Demagnetization of dolomitic shales, dolomites, and carbonate breccias of the Cambro-Ordovician Kap Stanton Formation within the Franklinian basin sequence of the North Greenland Fold Belt yields a characteristic direction (declination/inclination is 16/-16, tilt corrected) from 14 of the 28 sites. A negative breccia test and positive local and regional fold tests bracket the age of the magnetization as postdepositional but pre-Ellesmerian (pre-Carboniferous). After rotating the corresponding paleopole (1°N, 302°E) to account for the Cretaceous-Tertiary opening of the Labrador Sea, the Greenland pole remains removed by approximately 20° from the existing North American Apparent Polar Wander Path (APWP). However, the rotated paleopole (2°S, 284°E in North American coordinates) is similar to British Silurian and Devonian paleopoles in a Bullard et al. (1965) reconstruction (pole is 2°S, 322°E in British coordinates). We interpret our new paleopole as Late Silurian to Early Devonian in age, requiring a modification of previously published APWPs for Laurentia . If the age assignment is correct, the new paleopath for North America/Greenland and that for Great Britain can be brought into good agreement by closing the Atlantic Ocean, thus indicating that North America, Greenland, and Great Britain were assembled in a supercontinent in Late Silurian to Early Devonian time.

INTRODUCTION

Over the past 6 years there has been a major revision of the available North American Devonian paleomagnetic reference poles. Several paleopoles used for earlier paleogeographic reconstructions are now known to have been based on paleomagnetic data that were biased by Permo-Carboniferous remagnetizations [*Roy and Morris*, 1983; *Irving and Strong*, 1984, 1985; *Miller and Kent*, 1986 *a, b; Stearns and Van der Voo*, 1987]. Apparently reliable paleopoles for the Early Devonian of the Canadian Arctic [*Dankers*, 1982] and the Upper Devonian Catskill beds [*Miller and Kent*, 1986*a*, *b*] place North America in more southerly paleolatitudes in more recent Devonian paleogeographic reconstructions [*Van der Voo*, 1988]. Additionally, paleomagnetic data from the Traveler Felsite in Maine [*Spariosu and Kent*, 1983] and the Compton metasediments in Quebec [*Seguin et al.*, 1982] can be included in a list of reliable Devonian paleopoles, since the ages of these magnetizations are constrained by positive field tests, although it is not a priori certain that these areas belonged to the North American plate throughout the middle Paleozoic.

While progress has been made in the Devonian paleomagnetism of North America, the available data set is far from complete. Only five reliable paleopoles have been identified for all of North America. For the data from the Appalachian fold belt, the possibility of thrust sheet rotations causes some uncertainty about the paleomagnetic declinations but not the inclinations [*Miller and Kent*, 1986*b*]. Unquestionably, more detailed paleomagnetic work is needed in order to further constrain the Devonian paleogeography of North America.

In the remote north Greenland area, extensive paleomagnetic fieldwork has been undertaken by the Geological Survey of Greenland as summarized by *Abrahamsen and Marcussen* [1980], *Marcussen* [1981], and *Abrahamsen and Van der Voo* [1987*a*]. Until now, paleomagnetic results have been published from Proterozoic dikes [*Marcussen and Abrahamsen*, 1983; *Abrahamsen and Van der Voo*, 1987*b*] and Pliocence and Quaternary sediments [e.g. *Abrahamsen and Marcussen*, 1986], whereas no results of intervening ages have been published.

In this paper we present new paleomagnetic data from the Franklinian basin sequence in the North Greenland Fold Belt. Thermal demagnetization of samples from 14 sites in the Cambro-Ordovician Kap Stanton Formation [*Ineson and Peel*, 1989] yields characteristic magnetizations that appear to be Paleozoic in age. The age of the magnetization can be bracketed by a negative breccia test and positive local and regional fold tests. After comparison with published North American and European paleopoles, we interpret the results of this study as a new Laurentian paleopole of probable Siluro-Devonian age.

GEOLOGY AND SAMPLING

The geology of the northern coast of Greenland has been recently described in detail by several authors including *Soper and Higgins* [1987], *Surlyk and Hurst* [1983, 1984], *Hurst and Surlyk* [1983], and *Dawes and Kerr* [1982] and described in references within their works. A simplified geologic map is shown in Figure 1. Four east-west trending facies belts which have had different structural and environment histories are described by *Hurst and Surlyk* [1983]. These belts include a Precambrian crystalline basement in the south (covered by Proterozoic sediments and volcanics in the east), which is

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Paper number 89JB00583.
0148-0227/89/89JB-00583$05.00

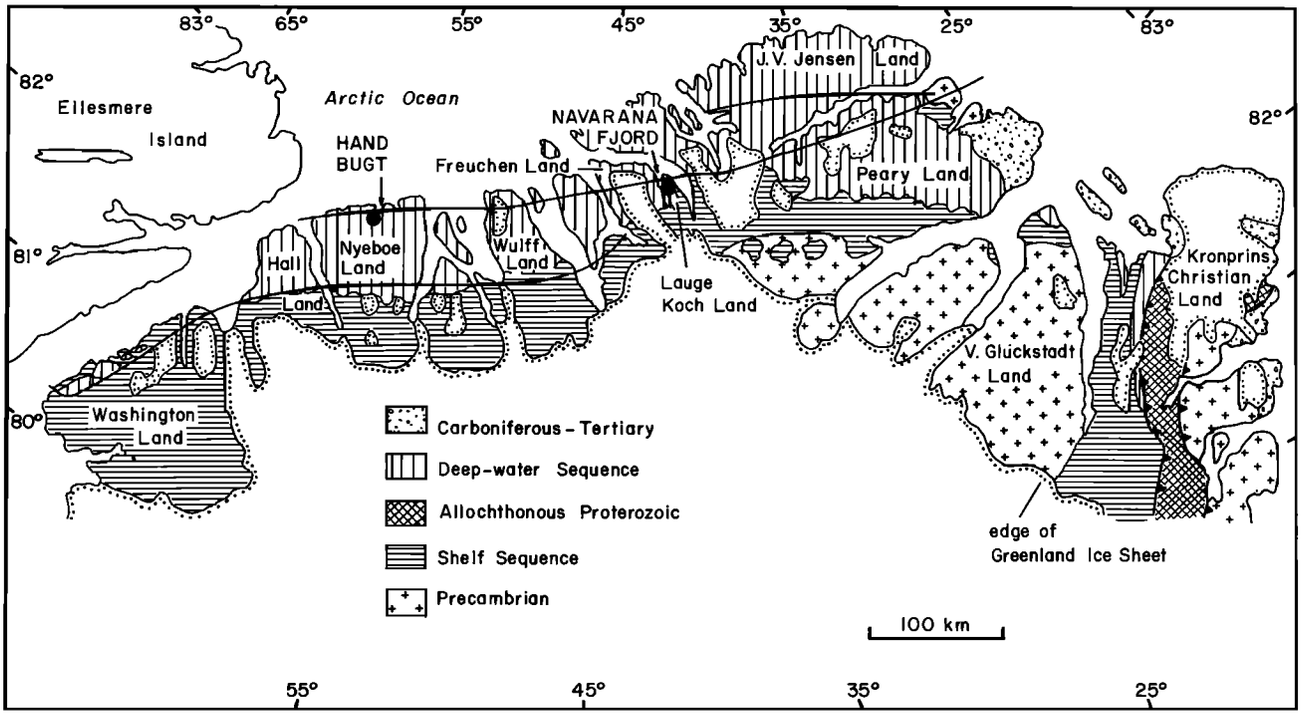


Fig. 1. Regional geologic map of northern Greenland with the two sampling localities at Hand Bugt and Navarana Fjord indicated as well as the main east-west lineaments mentioned in the text.

overlain to the north by a sequence dominated by shallow marine shelf carbonates, generally in platform facies. Continuing northward, one finds a shelf margin sequence whose position through time was controlled by a series of east-west trending faults and lineaments. Finally, to the north of the carbonate shelf margin was a deepwater basin characterized by the deposition of fine-grained terrigenous turbidites and occasional carbonate breccias derived from the shelf margin. The main part of this deeper-water carbonate sequence is Cambrian through Silurian in age [*Håkansson and Stemmerik*, 1984].

The deeper-water lower Paleozoic sediments were folded and lightly metamorphosed in the Devonian to Early Carboniferous Ellesmerian orogeny [*Dawes*, 1976; *Higgins et al.*, 1985]. In general, metamorphic grade is low and gradually increases northward. Approximately 200 km to the northeast of our sampling area, in the northernmost parts of Johannes V. Jensen Land, the deformation is intense, and the metamorphism locally reaches amphibolite grade. Rb-Sr dates on the metamorphics reported by *Springer* [1981] are Devono-Carboniferous. In easternmost Peary Land the deformed sequence of the North Greenland Fold Belt is unconformably overlain by a thick sequence of Upper Pennsylvanian and younger sediments [*Surlyk et al.*, 1980].

The overall tectonic setting of the carbonate basin is not well known, particularly since the northern side of the basin is not exposed. According to *Surlyk and Hurst* [1984], the possible tectonic interpretations include either a passive margin of a narrow ocean basin to the north or an ensialic deepwater basin formed by the rift-controlled subsidence of an area between the carbonate shelf to the south and a second landmass to the north. Similarly, the deformation of the basin could be the result of the closing of the narrow ocean basin or an ensialic orogeny.

We report here on new paleomagnetic results from the Cambro-Ordovician Kap Stanton Formation, collected from the two localities shown in Figure 1. In the central part of northern Nyeboe Land along the western edge of Hand Bugt, samples were collected from 18 sites in an overturned but homoclinal section (all strikes and dips are listed in Table 1). In Freuchen Land, 10 sites were collected in a broad, upright anticline exposed along Navarana Fjord. *Higgins and Soper* [1985] and *Jakobsen and Stendal* [1987] describe the lithologies of an unnamed slope sequence as shales of varying carbonate content, intraformational carbonate breccias interdigitated with thin-bedded dolomitic mudstones, shales, and dolomites (informally called the "tiger limestones"), black shales, and cherts. Most of the samples collected for this study were shaley dolomites, carbonate breccias and dark dolomitic shales from Unit 3 of their unnamed sequence, recently defined as the Kap Stanton Formation [*Ineson and Peel*, 1989].

Samples were collected in the field with a portable gasoline-powered drill and oriented with a sun compass. Magnetic compass readings were also taken but because of the extremely high latitudes of the sites the magnetic readings are thought to be less reliable than the sun compass measurements. Nevertheless, the magnetic readings serve as a useful albeit imprecise check on the sun compass orientations. At least three samples were collected per site, and these were later trimmed into standard 2.2 x 2.5 cm specimens. Natural remanent magnetizations (NRMs) were measured using an Superconducting Technologies cryogenic magnetometer

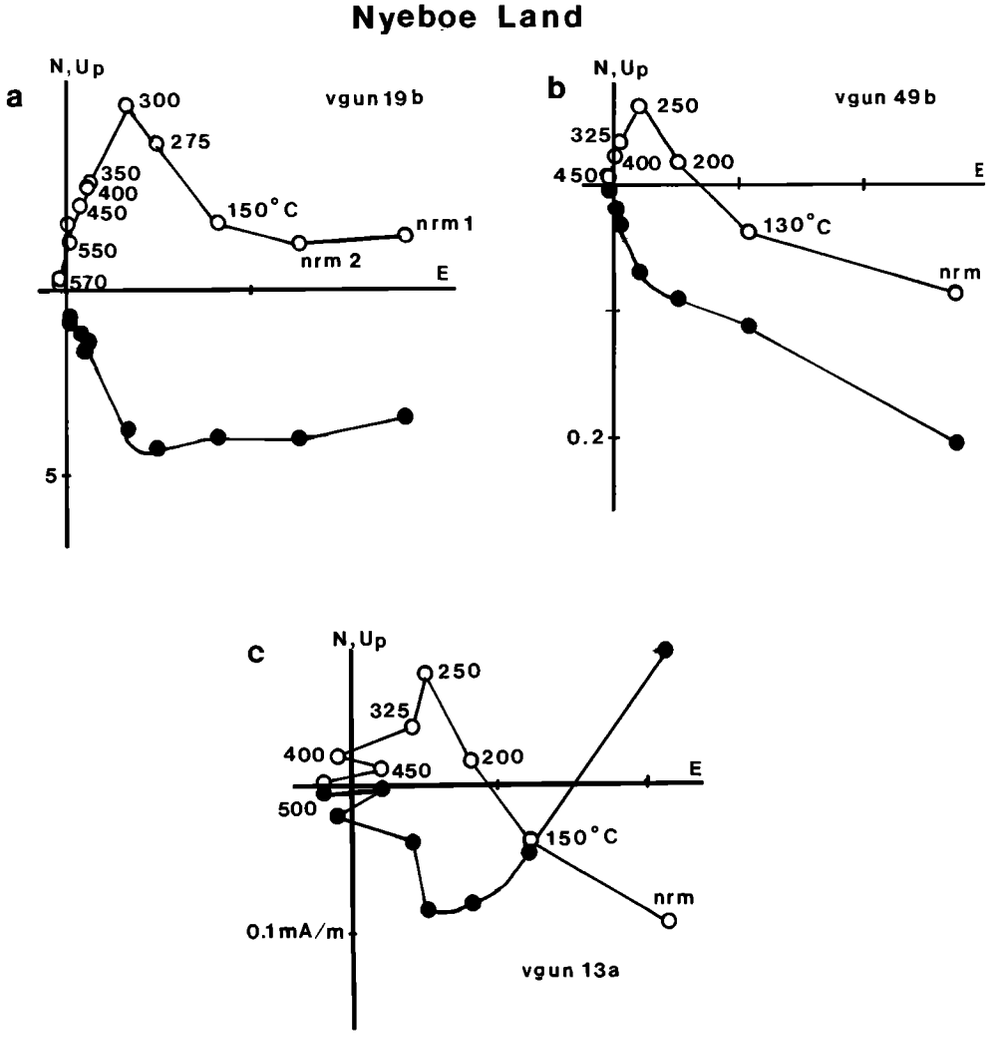


Fig. 2. Representative in situ vector endpoint diagrams for samples from Nyeboe Land: *(a)* and *(b)* characteristic southerly and up directions, and *(c)* representative "noisy" diagram typical of the less stable samples . Open (closed) symbols are projections on the vertical (horizontal) plane. Intensities are labeled on the axes in milliamperes per meter.

located at the University of Michigan. Samples were treated and stored in a magnetic field-free room in order to reduce the possible acquisition of viscous magnetizations. Standard thermal and alternating field demagnetization experiments were done using a Schonstedt Model TSD-1 thermal specimen demagnetizer and a Schonstedt GSD-1 alternating field demagnetizer. Comparison of the results of both thermal and AF demagnetization on specimens from the same sample showed that thermal demagnetization produced results that were less noisy and resolved components of magnetizations better. Therefore most of the samples were thermally demagnetized. After visual inspection of vector endpoint diagrams [*Zijderveld*, 1967] and stereoplots of the demagnetization trajectories, best-fit linear and planar segments were determined using principal components analysis [*Kirschvink*, 1980]. Only the results that passed standard statistical criteria (a minimum of three samples per site, and mean angular deviations of less than 12° for lines and planes) were used in calculating site means. To calculate a paleopole for the formation, we combined great-circle results with stable-endpoint results using the method of *Bailey and Halls* [1984].

PALEOMAGNETIC RESULTS

Nyeboe Land

Of the 18 sites that were sampled in Nyeboe Land, 10 sites revealed a characteristic southerly and up direction (in situ, mean is 162/-45) defined by at least three colinear points on a *Zijderveld* [1967] diagram. The typical in situ Zijderveld diagrams in Figures 2*a* and 2*b* show that after the removal of a low-temperature component the samples reveal a southerly and up direction above 250°C. Unblocking temperatures for this component ranged from 250° to 570°C which suggests that (titano-) magnetite may carry part or all of the NRM. The samples sometimes became magnetically unstable after 550°C treatments, so that the upper limit of the unblocking temperatures is not always precisely known. The characteristic sample directions for the 44 samples from these 10 sites are plotted in Figure 3*a*. In situ and tilt-corrected site means are plotted on equal-angle stereoplots in Figures 3*c* and 3*d* and are listed in Table 1.

In contrast to the stable magnetic behavior seen in 10 of the

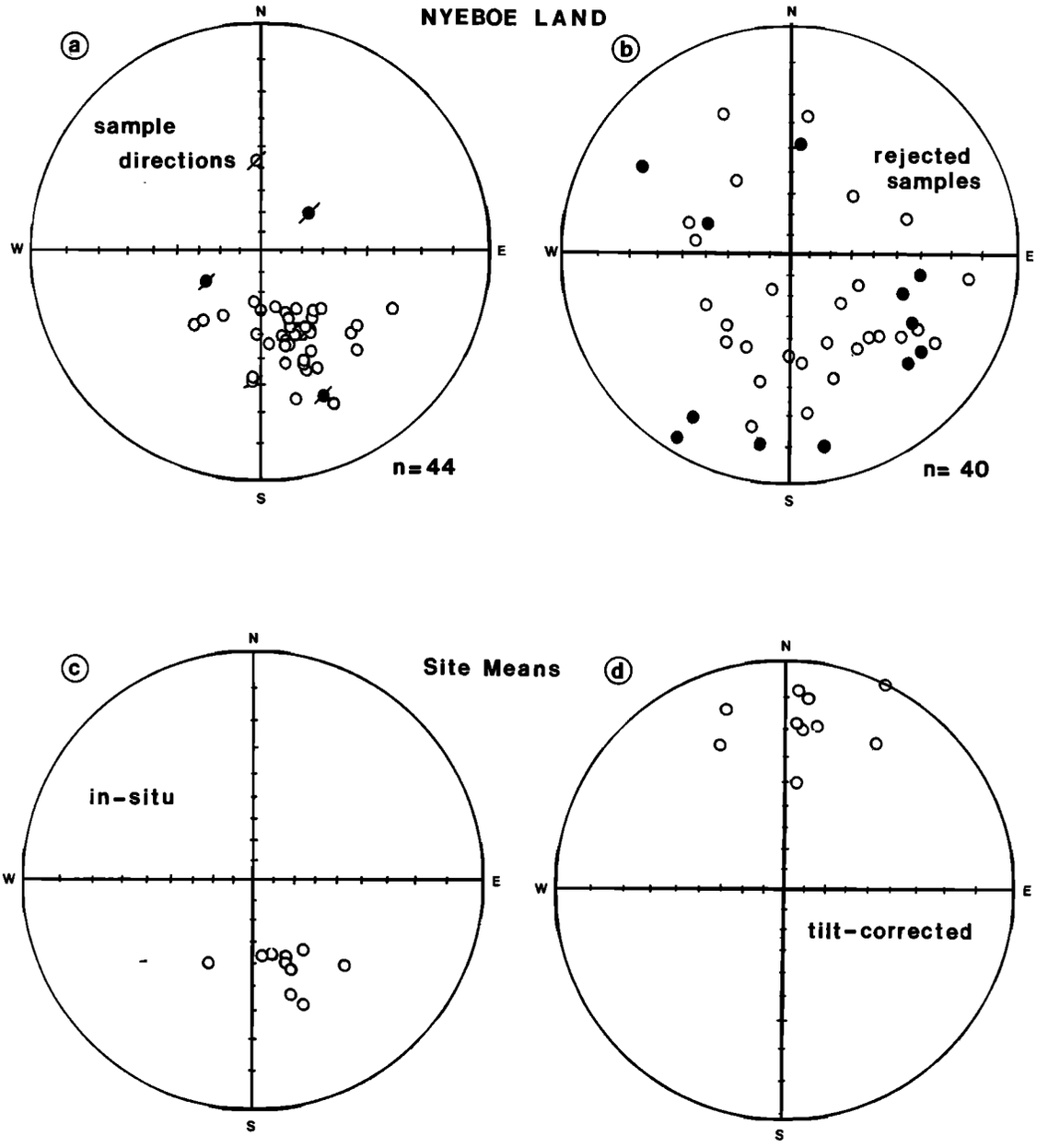


Fig. 3. Equal-angle stereoplots of the magnetic directions for the samples from Nyeboe Land: *(a)* directions calculated for individual samples from the well-behaved sites with slashes through the samples that were omitted from site mean calculations, *(b)* directions from the samples that were rejected,and *(c)* and *(d)* in situ and tilt-corrected site means, respectively. Open (closed) symbols are upper (lower) hemisphere projections of negative (positive) inclinations.

18 sites, data from the remaining eight sites were much noisier. Although southerly and up magnetizations were observed in many samples from the remaining eight sites, the within-site behavior was much less coherent. Characteristic directions for the 40 samples from these sites are plotted in Figure 3*b*. Many of the Zijderveld diagrams for these samples are like that shown in Figure 2*c*. This sample carries a southerly and up magnetization, but it is difficult to "pick" a direction on the Zijderveld diagram using appropriate statistical methods. These data do not pass our minimum quality criteria: for best-fit lines these include a minimum of three colinear points and MADs (mean angular deviations) less than 12°, whereas for calculations of site means we have not included the data from sites with fewer than three samples. In contrast, the MADs associated with the calculated directions from the 10 reliable sites are usually less than 5°. Furthermore, the results of planar analysis of the rejected demagnetization trajectories [*Kirschvink*, 1980] also produce statistically unacceptable results. Even though we have excluded the results from eight sites from the final results, the stereoplot in Figure 3*b* indicates that a majority of these samples also appears to carry magnetizations with southerly declinations and negative inclinations.

Sites 296 and 297 were collected in an intraformational breccia so that individual paleomagnetic samples contained both matrix and clasts. All samples from site 301 were taken

TABLE 1. Paleomagnetic Results for the Kap Stanton Formation of Northern Greenland

Site	s/d	n/N	D/I In Situ	D/I Tilt Corrected	k	α_{95}	GGU Ref. No.
			Hand Bugt, Nyeboe Land (82.1°N, 57.2°W)				
291	91/105	3/4	144/-49	32/-16	133	10.7	318601
292	87/110	4/4	157/-50	11/-18	460	4.3	318602
293	88/109	4/5	208/-45	336/-21	25	18.9	318603
294	87/109	4/4	166/-53	4/-18	57	12.2	318604
296	90/119	5/6	173/-53	4/-8	21	17.0	318606
297	71/120	4/4	159/-48	342/-11	103	9.1	318607
299	81/110	3/5	158/-29	6/-40	43	19.1	318609
300	84/125	4/4	157/-44	7/-10	157	7.3	318610
301	85/125	3/4	162/-34	6/-20	334	6.8	318611
302	83/140	4/4	133/-33	26/0	23	19.5	318612
Mean directions							
In situ for 10 sites			162/-45		24	10.0	
Tilt corrected for 10 sites				6/-17	18	11.8	
			Navarana Fjord (82.5°N, 42.36°W)				
307	225/21	5/5	350/7	350/-10	402	3.8	318617
309	232/16	3/4	355/21	353/7	137	10.6	318619
320	59/37	5/5	31/-14	28/5	230	5.1	318630
321	59/37	5/5	34/-11	33/6	77	10.5	318631
Mean directions							
In situ			13/-1		8	33.8	
Tilt corrected			11/2		12	28.2	
Site mean great-circle poles							
307		5/5	84/-15	86/1		18.9	
309		4/4	266/-2	264/-11		4.3	
320		5/5	121/-12	110/-43		8.3	
321		5/5	122/-9	112/-40		11.8	
Great-circle intersection (4/4 site mean poles)							
In situ			144/79		44	20.0	
Tilt corrected				358/-24	971	9.5	
Combined great-circle and stable endpoint mean*							
$N = 14$ site means				16/-16	16	11.1	

The parameters *s/d* are the strike and dip of the beds, *n/N* is the ratio of the samples used to samples analysed, *D/I* are the declination/inclination, and k and $\alpha 95$ are the precision parameter and the radius of the cone of 95% confidence, respectively [*Fisher*, 1953]. GGU Ref. No. is the Geological Survey of Greenland reference number.

*Tilt-corrected, in Navarana Fjord site coordinates. Corresponding paleopole: 1°N, 302°E, dp = 5.66, dm = 11.45. The parameters dp and dm are the semiaxes of the oval of 95% confidence about the pole position.

from a large, rotated block within one of the breccias. Since these three sites carry the characteristic magnetization direction, yet are from breccias, the breccia test is negative, and therefore the characteristic magnetization of the Kap Stanton Formation along Hand Bugt was acquired after the breccias were formed during Cambro-Ordovician deposition.

Navarana Fjord

The samples from these sites are thinly bedded dark shales. Four of the 10 sites, two from each limb of the broad, upright anticline, revealed characteristic northerly and shallow remanence directions. Typical Zijderveld diagrams for samples from these four sites are shown in Figures 4*a* and 4*b*. These samples showed the presence of two components of magnetization, (1) a steep downward direction removed at low temperatures that is likely to be an overprint related to the present-day field and (2) a characteristic, northerly and shallow component removed at higher temperatures. However, for reasons we will discuss below, we wish to raise the possibility that the steeply down overprint has blocking temperatures that overlap with those of the characteristic remanence component. While the Zijderveld plots (e.g., Figures 4*a* and 4*b*) appear to present relatively colinear segments, it may be possible that the apparently "stable" endpoints are still contaminated. We will return to this possibility after a discussion of the directions from the stable endpoints and the resulting fold test.

Characteristic sample directions for the 19 samples are shown in Figure 5*a*. Inspection of the in situ and tilt-corrected site means for the four sites plotted in Figures 5*c* and 5*d* shows that in general all of the samples reveal northerly and shallow directions. However, in detail the data from the two limbs of the fold are different. The in situ directions from the two sites on the southeastern limb are northeasterly and up and rotate to

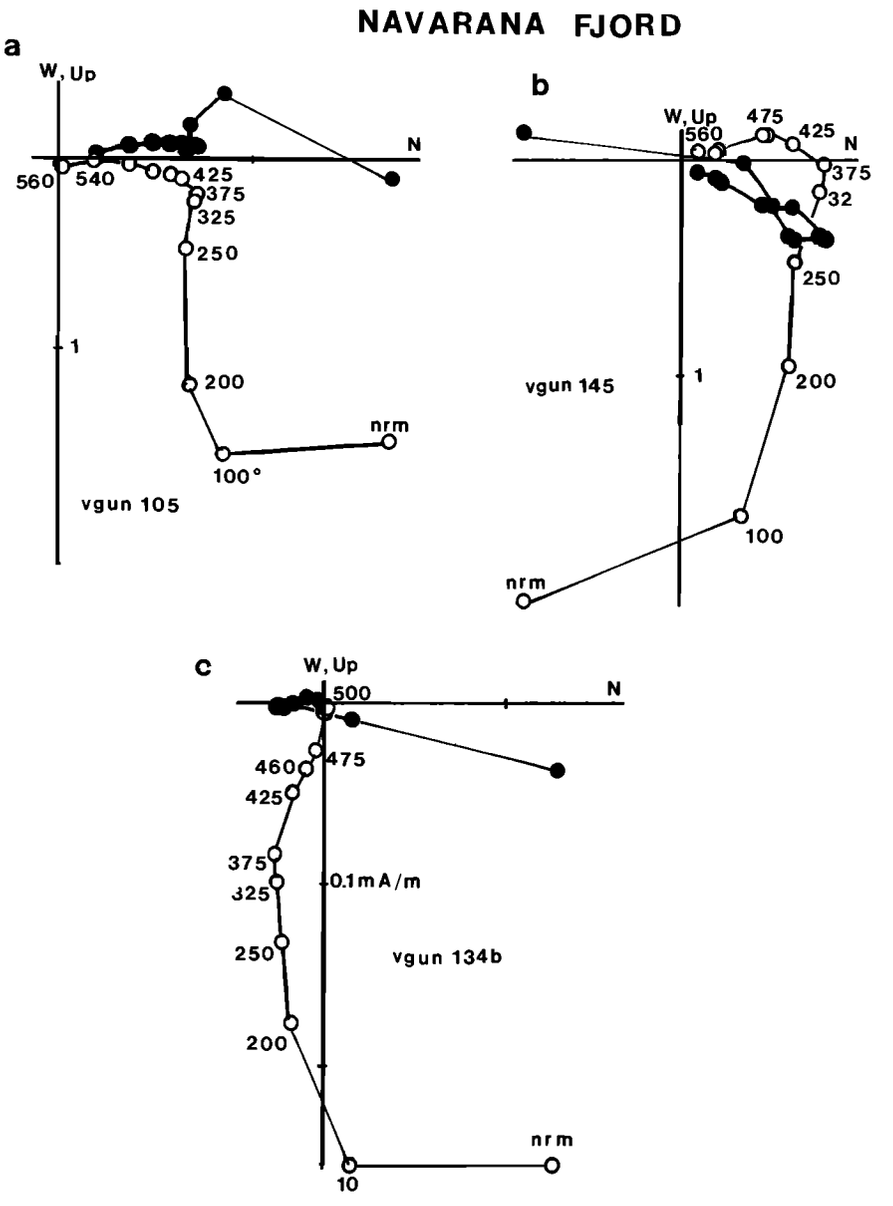


Fig. 4. Representative vector endpoint diagrams for the samples collected along Navarana Fjord. *(a)* and *(b)* northerly and shallow directions seen in four sites and *(c)* steeply downward direction typical of the remaining 6 sites. Symbols as in Figure 2.

northeasterly and down with the tilt correction, while the in situ directions from the two sites on the northwestern limb are north-northwesterly and down, and they become more shallow with the tilt-correction but stay north-northwesterly. In other words, the data are coherent from each limb, but application of the tilt correction neither increases nor decreases the dispersion of the data. The fold is not plunging, and the directions from the limbs differ in both declination and inclination; during unfolding the site means do not cross each other. Thus the characteristic directions determined from linear segments on Zijderveld diagrams do not make sense unless the remanence components were not completely resolved during demagnetization as already suggested above. We conclude that it is likely that the seemingly linear segments seen on the Zijderveld diagrams in Figures 4*a* and 4*b* are the result of the simultaneous removal of the high-temperature shallow northerly direction and a remnant of the lower-temperature steeply down direction. Separation of the two magnetic components during demagnetization is thought to have failed because of overlapping stability spectra.

To illustrate this further, demagnetization trajectories, both in situ and tilt corrected, for a representative sample from each limb are shown on the equal-angle stereoplots in Figure 6. For all 19 samples, the demagnetization trajectories follow great-circle paths, and this strengthens the suspicion that stable endpoints were not achieved during demagnetization. Individual trajectories for samples from the southeastern limb trend from southerly and down to northeasterly and shallow, whereas the data from samples from the northwestern limb trend northerly and shallow in a near-vertical plane. Great-circle analysis [*Halls*, 1976, *Kirschvink*, 1980] allows both the direction removed and the direction remaining to be determined because of the favorable geometry of the fold and the two components of magnetization. The results of the great-

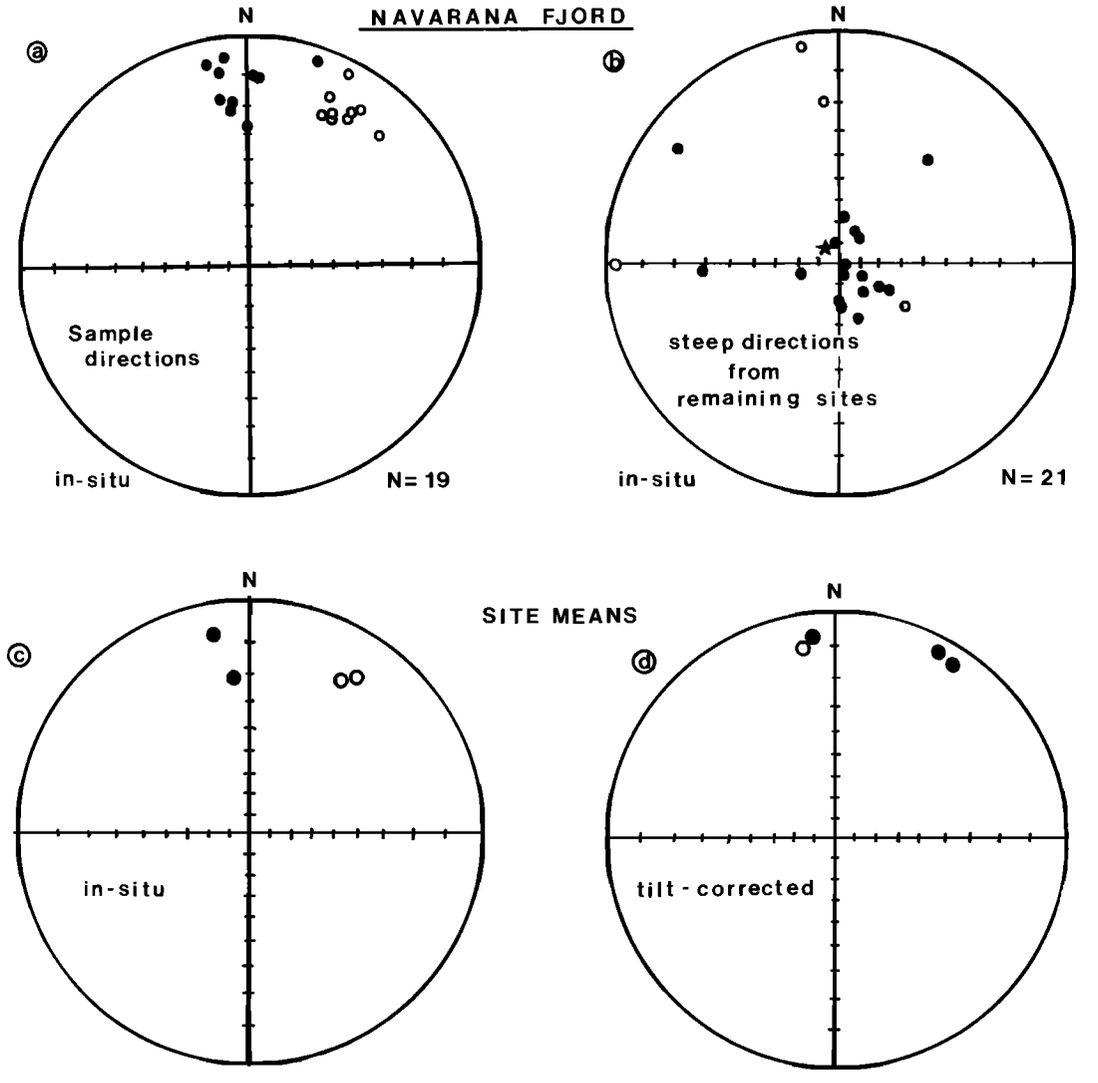


Fig. 5. *(a)* and *(b)* Equal-angle stereoplots of the characteristic sample directions , and *(c)* and *(d)* the site means for the samples from Navarana Fjord. The solid star in Figure 5*b* is the present-day dipole field for the site location at Navarana Fjord. Symbols as in Figure 3.

circle analysis are listed in Table 1, and the great-circle poles for the 19 samples as well as the four site means are plotted in Figure 7. To determine a mean direction, we calculated mean great-circle poles for each site and then calculated the intersection of the site means. Visual inspection of the stereoplots in Figure 6 confirms an in situ great-circle intersection at a southerly and down direction (144/79) and an intersection of the tilt-corrected great-circle paths at a northerly and up direction (358/-24). The southerly and steeply down direction is similar to the directions seen in samples from the other six sites as will be described below. Since the in situ intersection yields the lower-temperature component, this magnetization must be postfolding, whereas the tilt-corrected intersection, representing the high-temperature component, is prefolding. Thus the local fold test indicates that the northerly and up magnetic direction was acquired before the tilting of the rocks.

In contrast to the dark shale samples which reveal the shallow direction, the light-colored dolomite and carbonate breccia samples from the remaining six sites revealed mostly steeply downward directions of varying declinations (Figure 4*c*). This component has the same direction as the lower-temperature overprints in the samples of Figures 4*a* and 4*b*. These are plotted in Figure 5*b* along with the expected present-day field direction for north Greenland: these remanence components most likely record the recent geomagnetic field.

In summary, two magnetic directions are carried by the samples from Navarana Fjord. Samples from four of the 10 sites collected from the broad anticline yield ancient and shallow magnetizations. Dissimilarity of data from each limb of the fold, visual inspection of demagnetization trajectories, and great-circle analysis all indicate that stepwise demagnetization did not successfully separate the two magnetic components. The directions calculated from linear segments on Zijderveld diagrams are thought to be contaminated by the incomplete removal of a steeply downward direction. The direction of both components of magnetization, a prefolding northerly and up component and a postfolding southerly and steeply down component, has been determined by great-circle analysis. Most of the samples from the remaining six sites at

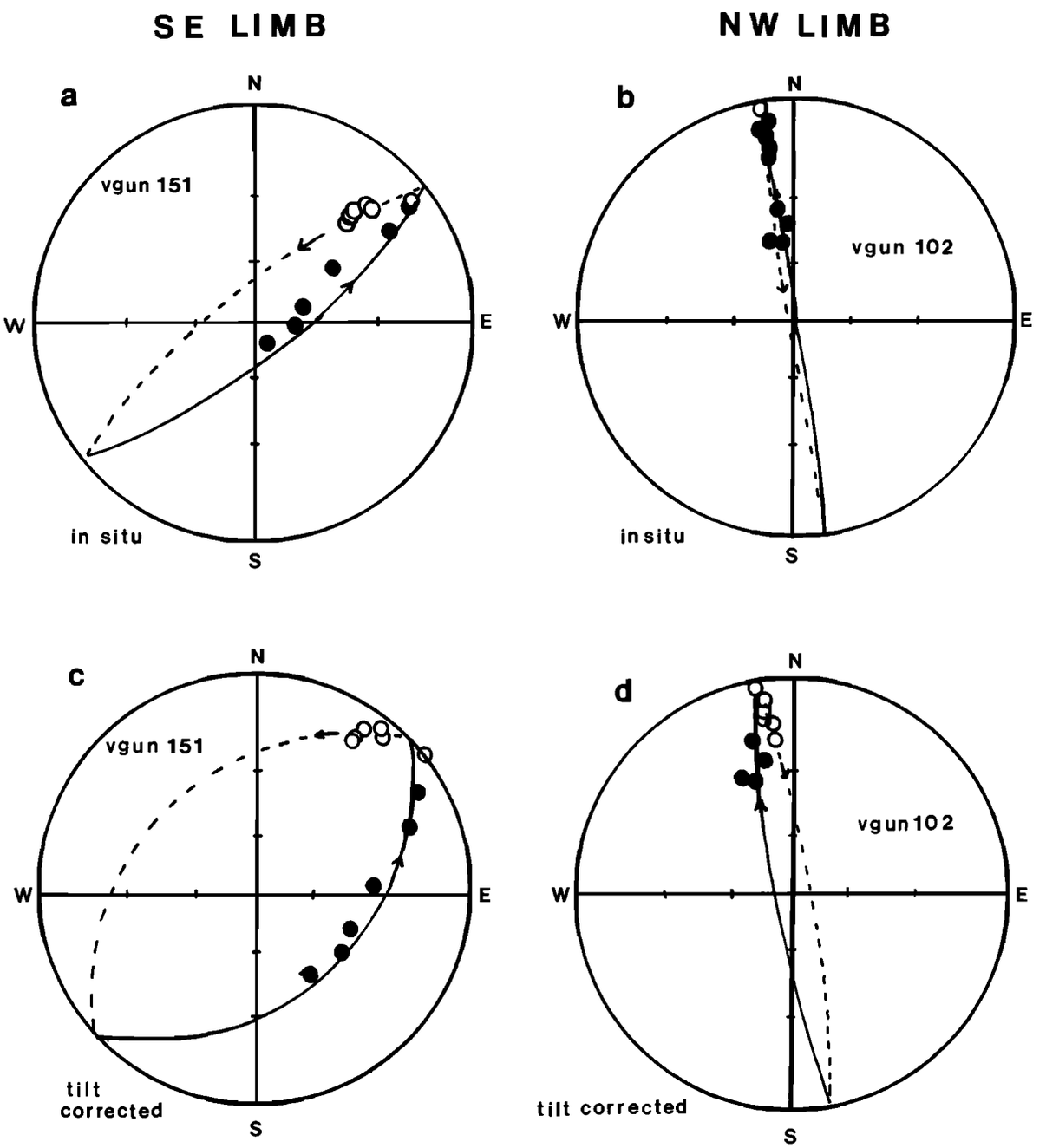


Fig. 6. Equal-angle stereoplots of the demagnetization trajectories for a typical sample from each limb of the anticline at Navarana Fjord. Best-fit planes are also shown. Symbols as in Figure 5.

Navarana Fjord reveal southerly and steeply downward directions interpreted as recent overprints.

Discussion of the Results

The samples from Nyeboe Land reveal a direction that is southerly and up (declination/inclination is 162/-45, $\alpha 95 = 10.0$) in situ and northerly, shallow and up (6/-17, $\alpha 95 = 11.8$) tilt corrected. A negative breccia test indicates that the magnetization is postdepositional. Since none of the sites sampled in slump folds revealed the characteristic magnetic direction and the section is homoclinal, no local fold test is possible, and the age of the magnetization cannot be further constrained without comparison with other data.

Great-circle analysis of the demagnetization data from four sites in a broad, upright anticline along the Navarana Fjord in Freuchen Land yields a prefolding northerly and up (358/-24, $\alpha 95 = 9.5$) direction and a post-folding steeply down (144/79, $\alpha 95 = 20.0$) direction. To compare the data from Hand Bugt with those from Navarana Fjord, paleopoles were calculated from the tilt-corrected site means for each of the 10 sites from Hand Bugt. We then calculated expected directions for the site coordinates at Navarana Fjord from each of those poles. These converted site mean directions are plotted in Figure 8 along

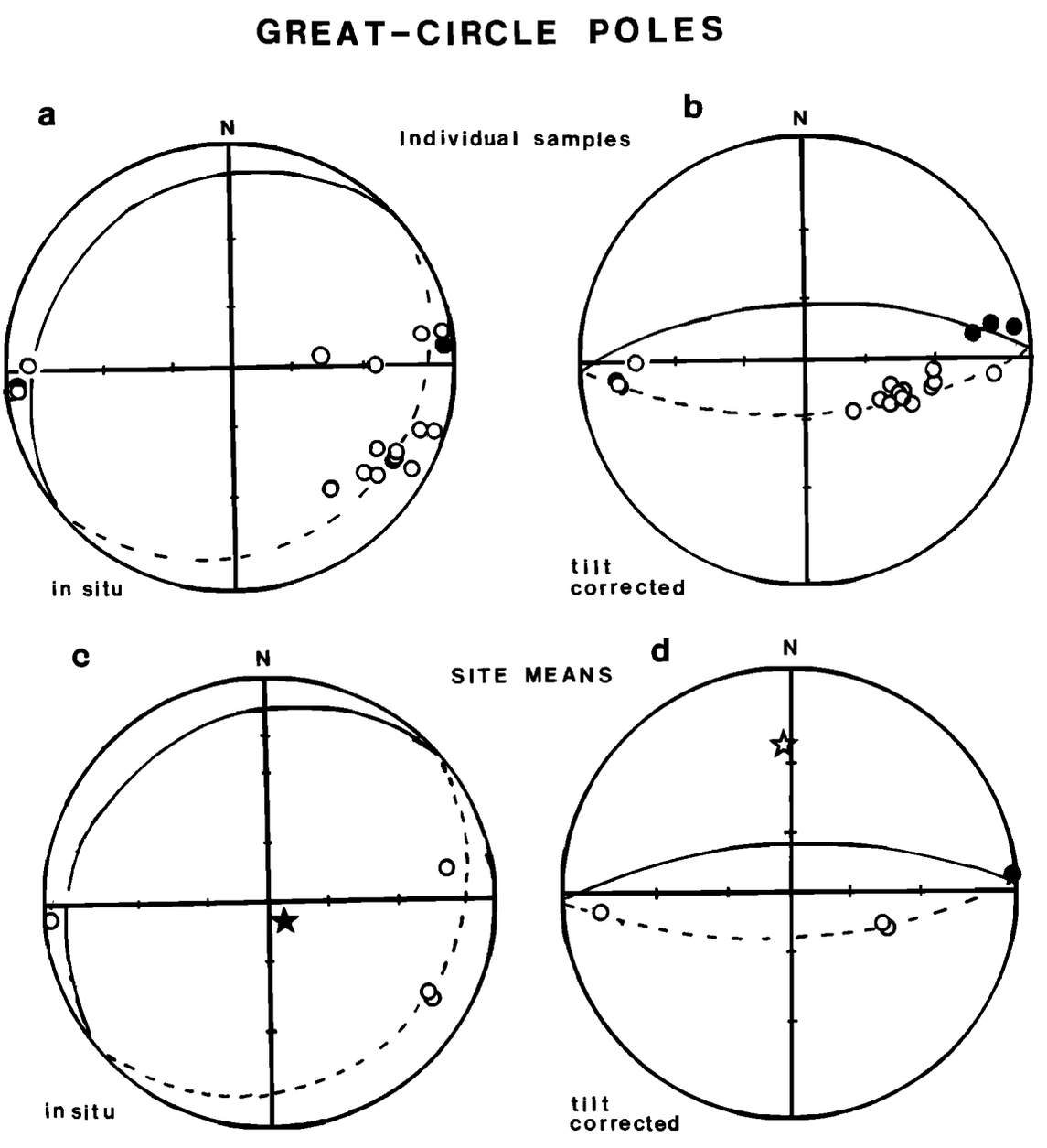


Figure 7. Great-circle poles for *(a)* and *(b)* individual samples and *(c)* and *(d)* site mean. Stars indicate the resulting great-circle intersections; symbols as in Figure 3.

with the great-circle results from Navarana Fjord. The direction corresponding to the tilt-corrected great-circle intersection is indicated by the star in Figure 8. The pole (358, -24) of this tilt-corrected great-circle result from Navarana Fjord falls within the group of tilt-corrected site means from Hand Bugt, whereas the in situ pole (144/79) is far removed from the recalculated in situ Hand Bugt directions. Thus the regional fold test is positive corroborating the conclusions derived from the local fold test. The age of the magnetization carried by the Cambro-Ordovician Kap Stanton Formation of North Greenland is therefore pre-Carboniferous (pre-Ellesmerian). To calculate a paleopole for the formation, great-circle site means were combined with the stable-endpoint site means using the method of *Bailey and Halls* [1984]. This paleopole is listed in Table 1.

The positive local and regional fold tests, the successful great-circle analysis, and the negative breccia test constrain the age of the magnetization of the Kap Stanton Formation to be post-Cambro-Ordovician (postdepositional) but pre-Carboniferous (prefolding). In an attempt to further constrain the age of the magnetization, we first compare our paleopole with previously published data from North America. In order to account for the Cretaceous-Tertiary opening of the Labrador Sea, we must rotate the Greenland paleopole with respect to the North American craton. We tried the rotation parameters of both *Srivastava* [1978] (11° of rotation about 71.5°N, 126.3°W) and *Bullard et al.* [1965] (19.42° of rotation about 65.8°N, 92.4°W) which yield rotated paleopoles for the Kap Stanton Formation of 2°S, 292°E, and 2°S, 284°E, respectively . No matter which rotation scheme is used, the rotated Greenland paleopole is

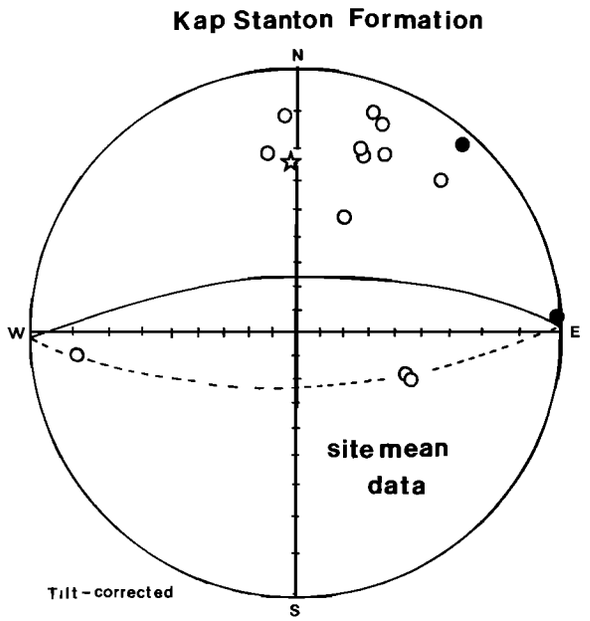


Fig. 8. Equal-angle stereoplots of the site mean data for the Cambro-Ordovician Kap Stanton Formation of North Greenland, plotted in the coordinates of the site at Navarana Fjord. The great-circle poles and corresponding intersection (star) for the four sites at Navarana Fjord (Figure 7*d*) were combined with the stable endpoint data from Nyeboe Land (Figure 3*d*, recalculated to the site of Navarana Fjord).

located approximately 20° from the North American Apparent Polar Wander Path (APWP) as is shown in Figure 9 where our result (Bullard reconstruction) is plotted with *Van der Voo*'s [1988] recently compiled North American APWP, a new paleopole for the Siluro-Devonian Andreas Formation [*Miller and Kent*, 1988], and a second pole for the Peel Sound Formation. It is highly unlikely that this discrepancy is the result of local tectonics or displacement of the entire North Greenland Fold Belt with respect to North America because the discrepancy is in paleolatitude; the sampling area would need to be located about 20° to the south of its present location with respect to North America, which is improbable given the cratonic setting of Greenland as part of Laurentia during the Paleozoic. We prefer, then, the interpretation that our result is a new North American post-Cambrian, pre-Carboniferous paleopole. Previously, *Dankers* [1982] reported two groups of magnetic directions carried by lower Devonian redbeds from the Canadian Archipelago. The corresponding paleopoles were 25°S, 279°E, and 1°S, 271°E. Although the equatorial paleopole (PS2 in Figure 9) is based on only 13 specimens from one site and therefore cannot be considered reliable, its similarity in paleolatitude to our result from northern Greenland lends credence to the possibility that our new result represents a Late Silurian to Early Devonian cratonic paleopole. The Andreas paleopole [*Miller and Kent*, 1988] is also of a lower latitude and further suggests that the Late Silurian to Early Devonian section of the North American

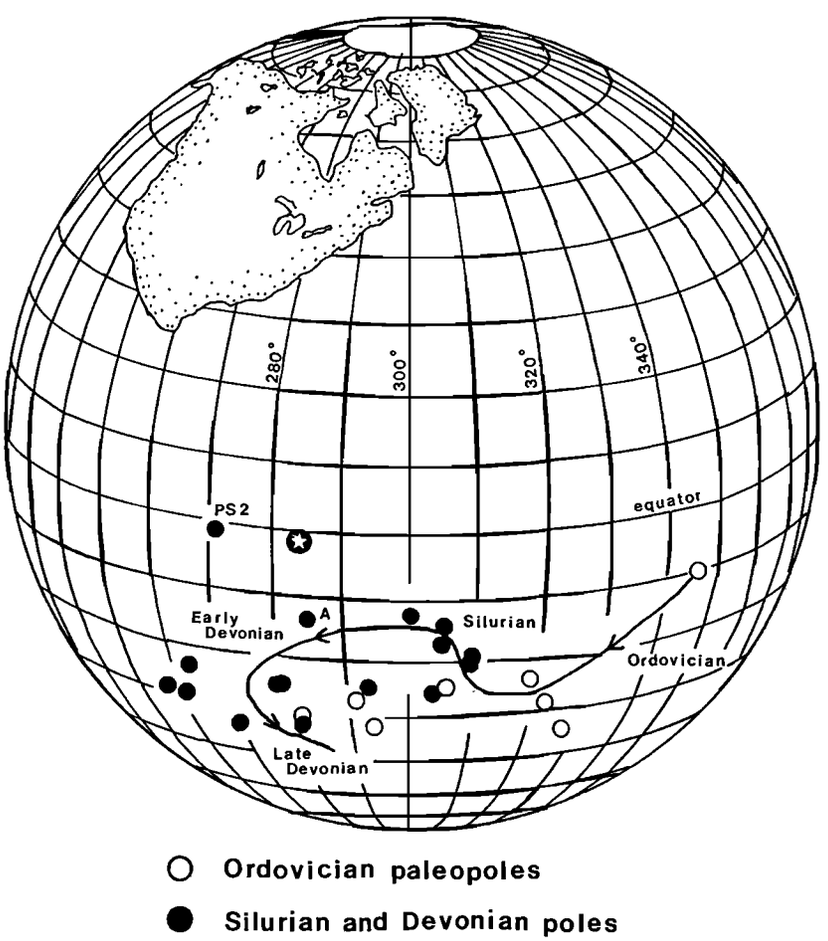


Fig. 9. *Van der Voo*'s [1988] middle Paleozoic Apparent Polar Wander Path for North America. Paleopole for this study (plotted in North American coordinates) is shown by the encircled star. PS2 and A are the poles for the Peel Sound and Andreas Formations, respectively. Open circles are Ordovician paleopoles, closed circles Silurian and Devonian poles.

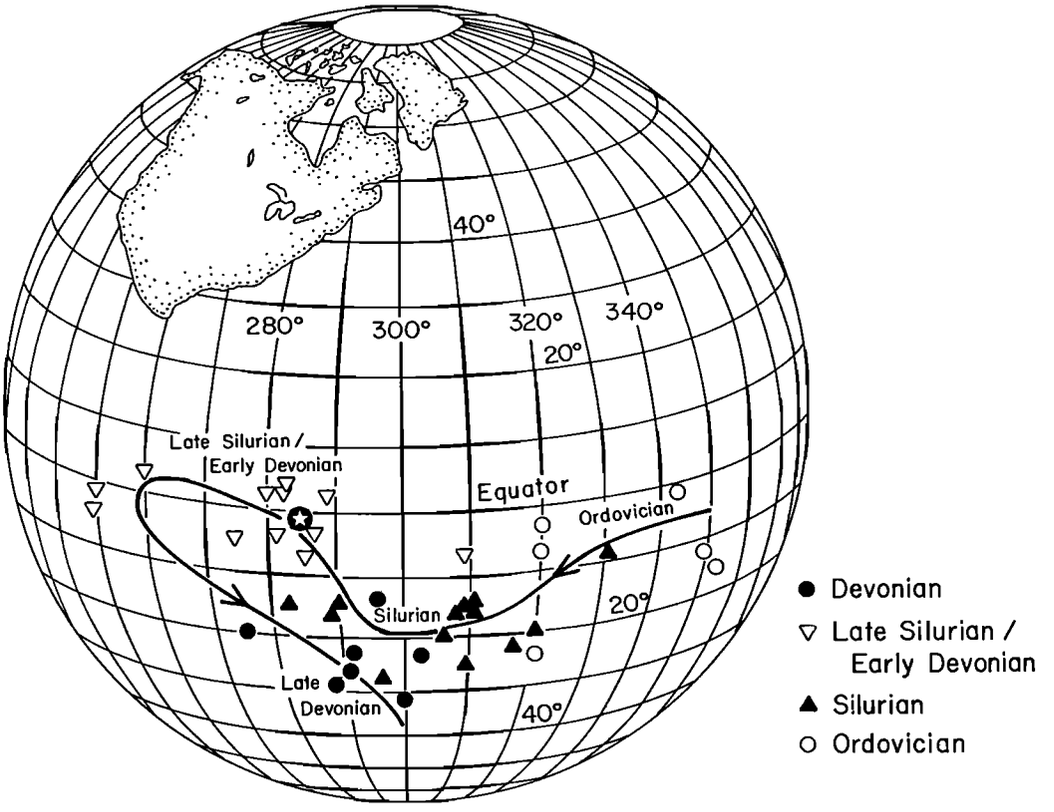


Fig. 10. Middle Paleozoic Apparent Polar Wander Path for Great Britain and Ireland using the paleopoles listed in Table 2 for Q = 3 or greater. The Devonian paleopoles are for all of Great Britain and Ireland whereas the Ordovician and Silurian paleopoles are from north of the Solway line (Iapetus suture) only. The paleopole for the Kap Stanton Formation is shown by the encircled star. All paleopoles have been rotated into North American coordinates according to a *Bullard et al.* [1965] reconstruction.

APWP lies in lower latitudes than previously thought. Currently, the paucity of reliable North American Late Silurian and Early Devonian paleopoles makes it difficult to further evaluate this interpretation of our Greenland paleopole.

A comparison of our paleopole with data from Great Britain, however, appears to give additional evidence for a Late Silurian - Early Devonian age. In Figure 10 we have plotted the available Ordovician, Silurian paleopoles from Britain and Ireland north of the Solway line (Caledonian or Iapetus suture) and Devonian paleopoles for all of Britain and Ireland (listed in Table 2) and our paleopole which has been rotated according to the *Bullard et al.* [1965] (38° rotation about 88.5°N, 27.7°E) reconstruction. The pole from the Kap Stanton Formation of North Greenland falls among the Silurian and Early Devonian poles. We interpret this as a strong indication that the Greenland paleopole is a new North American paleopole of Late Silurian to Early Devonian age. If this is true, then our result from the Kap Stanton Formation could be considered the first evidence that North America - Greenland and Britain were assembled in a *Bullard et al.* [1965] fit in the Late Silurian to Early Devonian. As is shown in Figure 11, the Greenland paleopole places North America and Greenland farther south in latest Silurian - earliest Devonian time than has been recently suggested [*Van der Voo*, 1988].

As a final note, the paleopole calculated for the in situ directions from Nyeboe Land (38°N, 133°E) is similar to previously published Late Carboniferous poles for North America. Although this permits the interpretation that the magnetization of the rocks in Nyeboe Land is a Kiaman overprint, the positive regional fold test indicates that the magnetization is prefolding. The remoteness of North Greenland from the Hercynian-Alleghenian orogenic belt makes it seem unlikely that these rocks would be affected by thermal events or fluids of a Permo-Carboniferous orogeny. Furthermore, a Permo-Carboniferous overprint age for the Nyeboe Land direction does not explain the paleopole for the results from the rocks collected from Navarana Fjord. There, successful great-circle analysis yields a positive local fold test and therefore brackets the age of magnetization as pre-Carboniferous; moreover, the Navarana Fjord directions (in situ or tilt corrected) do not match the directions expected for the late Paleozoic Kiaman interval.

CONCLUSIONS

A paleomagnetic investigation of the Cambro-Ordovician Kap Stanton Formation from northern Greenland has revealed a characteristic northerly and up-directed magnetization (mean declination/inclination is 16/-16, k=16.4, $\alpha 95$=11.1, tilt corrected) that fails a breccia test and passes local and regional fold tests. Thus the age of the magnetization can be bracketed as post-Cambrian and pre-Carboniferous. The corresponding paleopole (1°S, 122°E) remains removed by approximately 20° from the North American middle Paleozoic APWP after rotating the pole to account for the later opening of the Labrador Sea. We interpret the result as a new North American Siluro-Devonian paleopole. Comparison of this paleopole to British paleopoles rotated according to a *Bullard et al.* [1965]

TABLE 2. Devonian through Ordovician paleopoles from Britain and Ireland North of the Caledonian Suture

Rock Unit, Location	Age	Pole Position Lat, Long	k	$\alpha 95$	Reliability 1	2	3	4	5	6	7	Q	Reference
Portishead Redbeds, West England	Du/Cl	32N, 158E	23	10	x	x		x	x			4	*Morris et al.* [1973]
Orkney lavas	Dm/Du	24N, 150E	45	9	x				x		x	3	*Storetvedt and Petersen* [1972]
Esha Ness, Scotland	Dm/Du	21N, 134E	58	3	x	x	x		x	x	x	6	*Storetvedt and Torsvik* [1985]
Foyers Old Red Sandstone, Scotland	Dm	30N, 147E	8	29	x				x		x	3	*Kneen* [1973]
Lower Caithness Redbeds, Scotland	Dm	27N, 149E	20	6	x		x		x	x		4	*Storetvedt and Torsvik* [1983]
Sarclet Sandstone, Scotland	Dl/Dm	24N, 161E	14	6	x	x	x		x	x		5	*Storhaug and Storetvedt* [1985]
Hendre, Blodwell Intrusives,Welsh Borderlands	384	32N, 166E	21	15	x	x						2	*Piper* [1978a]
Hoy lavas, Scotland	Dl	14N, 154E	66	4	x		x		x		x	4	*Storetvedt and Meland* [1985]
Eday Sandstone, Scotland	Su/Dl	8N, 167E	8	8	x		x	x	x		x	5	*Robinson* [1985]
Orkney lavas	Su/Dl	2S, 147E	18	14	x	x			x	x	x	5	*Morris et al.* [1973]
Cheviot hills, combined, Midland Valley	398	4S, 141E	11	13	x	x			x	x	x	5	*Thorning* [1974]
Old Red Sandstone, Anglo-Welsh Cuvette	Su/Dl	3S, 118E	5	13	x		x	x	x		x	5	*Chamalaun and Creer* [1964]
Moine metasediments, Scotland	Su-Dl	6N, 133E	74	6		x	x			x	x	4	*Watts* [1982]
Garabal Hill-Glen Fyne, Scotland	404	5N, 146E	11	24	x		x				x	3	*Briden* [1970]
Lorne Plateau Lavas, Midland Valley	405	2S, 141E	43	6	x	x		x	x		x	5	*Latham and Briden* [1975]
Comrie Intrusion, Scotland	408	6N, 107E	19	6	x	x	x				x	4	*Turnell* [1985]
Strathmore Lavas, Midland Valley	408	2S, 138E	18	3	x	x	x	x	x	x	x	7	*Torsvik* [1985a]
Midland Valley Lavas	408	4N, 140E	13	6	x	x		x	x	x	x	6	*Sallomy and Piper* [1973]
Salrock Formation, Northwest Ireland	Sm/u	2N, 108E	44	4	x	x	x	x		x	x	6	*Smethurst and Briden* [1988]
Arrochar Igneous Complex, Scotland	414	8N, 144E	167	5	x	x	x	x			x	5	*Briden* [1970]
Aberdeenshire Gabbros III, Scotland	S	5N, 151E	621	5			x				x	2	*Watts and Briden* [1984]
Ratagan Intrusion, Scotland	415	15N, 167E	30	8	x	x	x	x		x	x	6	*Turnell* [1985]
Clare Island, Northwest Ireland	Sm	28N, 154E	27	10	x	x	x	x			x	5	*Smethurst and Briden* [1988]
Peterhead Granite, Scotland	Sl-m	21N, 178E	19	5		x	x				x	3	*Torsvik* [1985b]
Foyers Plutonic Complex, Scotland	Sl-m	16N, 168E	6	19			x			x		2	*Kneen* [1973]
Foyers Granite, Scotland	Sl-m	27N, 167E	15	6		x	x				x	3	*Torsvik* [1984]
Helmsdale Granite, Scotland	Sl-m	23N, 175E	32	6		x	x				x	3	*Torsvik et al.* [1983]
Borrolan Syenite, Scotland	S	18N, 147E	11	8		x	x			?	x	3	*Turnell and Briden* [1983]
N. Highlands Zone 5, Scotland	S?	16N, 168E	14	6		x	x			x	x	4	*Esang and Piper* [1984]
Loch Ailish, Scotland	S	16N, 141E	14	8		x	x				x	3	*Turnell and Briden* [1983]
Assynt Dikes Combined, Scotland	S	16N, 148E	32	5		x	x				x	3	*Turnell and Briden* [1983]
Strontian Granite, Scotland	Sl-m	21N, 164E	18	5		x	x				x	3	*Torsvik* [1984]
Borrolan Ledmorite, Scotland	430	10N, 189E	13	7	x	x	x			x	x	5	*Turnell and Briden* [1983]
Caledonian Dolerites, Scotland	435	15N, 168E	10	10	x	x	x			x	x	5	*Esang and Piper* [1984]
Caledonian Microdiorites, Scotland	435	17N, 166E	14	6	x	x	x			x	x	5	*Esang and Piper* [1984]
Tourmakeady, Glensaul, Ireland	Ou-S	31N, 169E	38	4		x	x					2	*Deutsch and Storetvedt* [1988]
Borrolan Pseudoleucite, Scotland	Ou-Sl	24N, 178E	16	8			x			x	x	3	*Turnell and Briden* [1983]
Ben Loyal, Scotland	Ou-Sl	4N, 178E	7	16			x			x	x	3	*Turnell and Briden* [1983]
Aberdeenshire Gabbros II, Scotland	447	8N, 178E	26	14	x	x	x				x	4	*Watts and Briden* [1984]
Barrovian of Angus, Scotland	Om-u	3N, 202E	47	7		x	x	x			x	4	*Watts* [1985]
Port Askaig Tillite Overprint, Scotland	O	28N, 198E	4	12			x				x	2	*Stupavsky et al.* [1982]
Ballantrae Intrusives Combined, Scotland	Om?	12N, 207E	24	8		x	x				x	3	*Piper* [1978b]
Aberdeenshire Gabbros I, Scotland	Ol-m	16N, 212E	121	8		x	x				x	3	*Watts and Briden* [1984]

Northern British Caledonide possibly displaced, rotated terranes (not used)

Mweelrea Ignimbrites, Ireland	Om	9N, 214E	23	20	x	x	x			x	3	*Deutsch and Somayajulu* [1970]
Slockenray Formation, Scotland	O/m	12S, 120E		4	x	x	x	x		x	5	*Trench et al.* [1988]

The entries in this table are all for Britain and Ireland for post-Silurian time and for Britain and Ireland north of the Solway line (Caledonian or Iapetus suture) for Silurian and Ordovician time. The reliability criteria (1–7), on which the quality factor *Q* is based, are as follows: (1) rocks are well dated, to within half a period, (2) at least 25 individually oriented sample directions have been used to calculate the mean, which is associated with *k* greater than 10 and a95 less than 16 degrees, (3) all entries are based on demagnetization studies, but this criterion is met if combined stereonets and intensity decay plots, and/or Zijderveld diagrams, and/or principle component analysis have been used and published in sufficient detail, (4) fold, conglomerate, or contact tests are available to constrain the age of the magnetization, (5) results are based on structural corrections and no suspicions exist about local rotations (for pre-Devonian time, this criterion is generally not being met), (6) reversals are present, and (7) the paleopole does not resemble those for younger rocks; an "x" mark under any of the seven reliability columns means that this particular criterion is being met. D is Devonian, S is Silurian, O is Ordovician, u is upper, m is middle, l is lower, Lat is latitude, Long is Longitude (in degrees), *k* and a95 are the precision parameter and radius of the cone of 95% confidence, respectively [*Fisher*, 1953]. A hyphenated age designation means that the best age estimate spans the range indicated (e.g., Sl-m means the entire lower and middle Silurian), whereas a slash implies that the best age estimate spans the boundary indicated.

Fig. 11. Paleogeographic reconstruction for latest Silurian–early Devonian time, with North America and Greenland positioned according to the new paleopole of this study. Other continental blocks are positioned following *Van der Voo* [1988].

reconstruction further suggests that the pole is Siluro-Devonian in age. Since the paleopole is similar to the previously published British Siluro-Devonian paleopoles, it may constitute the first paleomagnetic evidence that North America, Greenland, and Great Britain were assembled to form the supercontinent of Bullard et al. in the Late Silurian to Early Devonian.

Acknowledgments. This research was conducted under the joint auspices of the Geological Survey of Greenland (GGU) 1985 expedition, the Division of Earth Sciences, the National Science Foundation, grant EAR 84-07007 and the Danish Natural Science Research Council , grant 911-4711. We thank Niels Henriksen for his impeccable organization of the GGU expedition and for making the fieldwork (carried out by R.V. and N.A.) a truly unforgettable experience. This paper is published with the permission of the Director of the Geological Survey of Greenland.

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(Received October 28, 1988;
revised March 7, 1989;
accepted March 20, 1989)